1687 Isaac Newton discovered them in his own theory of gravity and used them to explain ocean tides.

The Moon's gravity pulls more strongly on the near face of the Earth than on the far face, Newton reasoned. And the direction of pull on the Earth's sides is slightly inward, because it is toward the Moon's center, a slightly different direction on the Earth's two sides. This is the usual viewpoint about the Moon's gravity depicted in Figure 4.8.

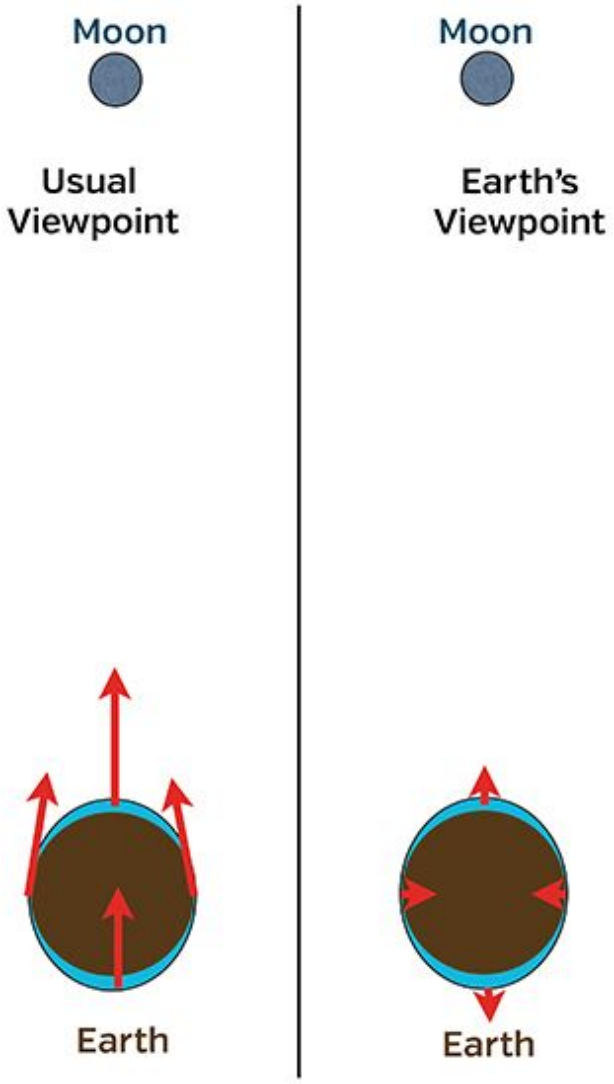


Fig. 4.8. Newton's explanation for the tides on the Earth's oceans.